

Technology Stack

Date	04 July 2026
Team ID	SWTID-2026-5594
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML5, CSS3, Bootstrap 5, JavaScript</i>	<i>Used to create a responsive, user-friendly, and interactive web interface for entering applicant details and displaying prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask handles HTTP requests, processes user input, loads the trained machine learning model, and returns prediction</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset (Application Record & Credit Record), Joblib (.pkl)</i>	<i>CSV files are used for model training, while the trained Random Forest model is stored as a .pkl file using Joblib for fast prediction.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Watson Machine Learning, Flask Local Server</i>	<i>IBM Watson Machine Learning is used for cloud deployment, while Flask is used for local testing and demonstration.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Git tracks source code changes, and GitHub enables version control,</i>

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Joblib, Jupyter Notebook, VS Code</i>	<i>Used for data preprocessing, visualization, machine learning model training, model evaluation, saving the trained model, and application development.</i>
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